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TrustLayer

Direct Lending Protocol

Smart contract mechanics. Loan structure and tranches.
Payment waterfall. Default scenario. Legal framework.

Prepared for professionals with financial industry background.

Blockchain concepts explained through banking analogies.

01

What Is a Smart Contract

A smart contract is code deployed on a decentralised computing network. It executes automatically when the conditions written at creation are met. No party can change the rules after the contract is recorded on-chain.

Banking analogy

The closest banking equivalent is a documentary letter of credit. A bank holds the buyer's funds and releases them to the seller upon presentation of shipping documents. A smart contract works the same way, but without the bank: the guarantor role is played by mathematical code recorded in a public ledger maintained by thousands of independent nodes.

If an L/C is a bank's promise, a smart contract is a mechanism with no owner that needs no instructions. Condition met - outcome delivered automatically.

Feature	Bank Contract	Smart Contract
Storage of terms	Bank server, closed system	Thousands of independent nodes simultaneously
Execution	Manager decides manually	Automatic, ~15 seconds
Changing terms	Possible via amendment	Impossible after on-chain recording
Status visibility	On request, business hours	Public, real-time, 24/7
Infrastructure failure	Delayed or failed payment	Network continues independently
Intermediary	Bank as mandatory participant	None

Three fundamental properties

Immutability. Once recorded on the blockchain, no party can alter the contract terms - not the platform, borrower, investor, or court order.

Automatic execution. The contract does not wait for anyone to trigger a payment. Maturity reached - funds distributed. Third missed payment - default procedure launched.

Transparency. Any market participant can check the contract state at any time: funds raised, token holders, missed payments, next payment date.

02

Master Contract and Tokens

The master contract is a software container holding all loan logic: amount, term, rate, tranche structure, payment rules, and default mechanism. Inside it are tokens - digital debt claims of fixed denomination.

A token certifies the holder's right to receive a proportional share of principal and accrued interest at the contract's maturity date. Legally the token is a Digital Financial Asset (DFA) under Federal Law 259-FZ.

Tranches: risk structuring

The master contract issues three categories of tokens - tranches - differentiated by payment priority and risk level. This model is identical to the tranche structure of corporate CLOs used in international banking.

Tranche	Share	Rate	Investor	Paid	Losses
A Senior	70%	12% p.a.	Retail investors	1st	Last
B Mezzanine	20%	18% p.a.	Risk investors	2nd	Middle
C Junior	10%	25% p.a.	Borrower only	Last	First

Activation mechanism: mandatory borrower contribution

Before the contract opens to external investors, the borrower must fully subscribe Tranche C - their own capital. The contract technically blocks purchase of Tranches A and B until Tranche C is fully funded. This is a code condition, not an administrative restriction.

A borrower who has put in their own money - which they receive last and lose first - has a direct financial incentive to service the debt. The banking analogy is a mortgage down-payment requirement.

Secondary market

Tranche A and B tokens are transferable securities. An investor needing liquidity before maturity can list tokens on the platform's secondary market. The master contract automatically records ownership changes and routes all future payments to the new holder. Tranche C tokens are non-transferable.

03

Full Loan Lifecycle

Ideal scenario: the borrower meets all obligations on time. All calculations verified - contract balance nets to zero.

Parameter	Value
Loan amount	RUB 1,000,000
Term	12 months
Borrower rate	20% p.a.
Platform origination fee	2% = RUB 20,000
Borrower corporate tax	20% (LLC, standard system)
Investor personal income tax	13% on interest income
Platform simplified tax	6% on revenue

Step 1. Contract issuance and fundraising

Investor	Tranche	Amount, RUB	Rate	Tranche share
Ivan	A Senior	200,000	12%	28.6%
Maria	A Senior	300,000	12%	42.8%
Peter	A Senior	200,000	12%	28.6%
Alexey	B Mezzanine	200,000	18%	100.0%
Borrower	C Junior	100,000	25%	100.0%
Total	-	1,000,000	-	-

Platform deducts RUB 20,000 fee. Borrower receives RUB 980,000 to their account.

Step 2. Business operations during the year

Line item	Amount, RUB
Annual revenue	2,200,000
Operating expenses	980,000
Loan interest (tax-deductible expense)	200,000
Pre-tax profit	1,020,000

Line item	Amount, RUB
Corporate income tax 20%	204,000
Net profit	816,000

Step 3. Loan repayment

Item	Amount, RUB
Principal	1,000,000
Interest 20% p.a.	200,000
Total into contract	1,200,000

Step 4. Automatic distribution of funds

Investor	Tranche	Principal	Income	Tax 13%	Net received
Ivan	A Senior	200,000	24,000	-3,120	220,880
Maria	A Senior	300,000	36,000	-4,680	331,320
Peter	A Senior	200,000	24,000	-3,120	220,880
Alexey	B Mezzanine	200,000	36,000	-4,680	231,320
Borrower	C Junior	100,000	25,000	-3,250	121,750

Verification: in RUB 1,200,000 = out to investors RUB 1,145,000 + platform spread RUB 55,000. Discrepancy: RUB 0.

04

Platform Economics

The platform earns from two revenue sources per transaction, taking no credit risk onto its balance sheet.

Source	Calculation	Amount, RUB
Origination fee (2%)	1,000,000 x 2%	20,000
Interest spread	Borrower rate 20% - avg investor rate 14.5%	55,000
Gross total	-	75,000
Simplified tax 6%	75,000 x 6%	-4,500
Net income per deal	-	70,500

Summary balance for all participants

Participant	Outcome	vs. Bank
Borrower	Effective rate ~20.2%. Net borrowing cost RUB 198,250.	Bank loan 27%. Saving ~RUB 68,000.
Tranche A investor	12% net p.a. Net income RUB 84,000.	Bank deposit 8% net. Income +81%.
Tranche B investor	18% net p.a. Risk transparent and numerically defined.	Higher yield with conscious risk.
Platform	Net income RUB 70,500 per deal.	Margin 7.5% vs bank's 17-22%.
Government (tax)	PIT + simplified tax + corporate tax = RUB 227,350.	Full transparency. All declared.

05

Default Mechanism

Default is the borrower's failure to meet payment obligations. The contract responds in steps: first applying financial pressure via penalty rates, then declaring default and distributing available funds. All automatic.

Stepped response to missed payments

Event	Contract action	Rate
Months 1-6: on time	Normal mode	20%
Month 7: first missed payment	Automatic rate increase. All investors notified.	24%
Month 8: second in a row	Rate increase. Default warning issued.	28%
Month 9: third in a row	Default declared. Tranche C frozen. Distribution triggered.	Default

Example: borrower returned 40% of principal

At month nine the business ceased operations. The borrower transferred RUB 400,000. Tranche C (RUB 100,000) has been locked since day one. Total available for distribution: RUB 500,000.

Tranche	Invested	Received	Loss	Loss %
A Senior	700,000	500,000	200,000	28.6%
B Mezzanine	200,000	0	200,000	100.0%
C Junior	100,000	0	100,000	100.0%
Total	1,000,000	500,000	500,000	50.0%

Value of borrower collateral

Scenario	Distribution pool	Tranche A loss	Loss %
Without borrower collateral	400,000	300,000	42.9%
With collateral (current model)	500,000	200,000	28.6%
Effect of collateral	+100,000	-100,000	-14.3 pp

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Diversification via Mixed Packages

Investors are not limited to a single borrower. The platform allows building token packages from multiple master contracts - analogous to a debt mutual fund, but without a management company or its fees.

Loan	Sector	Token amount	Rate	Weight
Borrower A	Manufacturing (food industry)	RUB 50,000	12%	33.3%
Borrower B	Retail trade	RUB 50,000	13%	33.3%
Borrower C	Logistics	RUB 50,000	11%	33.3%
Total	-	RUB 150,000	12%	100%

If one of three borrowers defaults, the investor loses no more than a third of their capital. Total portfolio loss requires simultaneous bankruptcy across different sectors - statistically improbable.

07

Legal Framework

The model operates within current Russian legislation without using cryptocurrencies or foreign jurisdictions.

Instrument	Legal basis	Description
Digital Financial Asset (DFA)	Federal Law 259-FZ, 31.07.2020	Digital claim right issued through a licensed information system. Smart contract has legal force.
Information System Operator	FZ-259, Bank of Russia licence	Entity licensed by the Central Bank to issue DFAs. Platform operates as ISO.
Digital Ruble	Federal Law 340-FZ, 24.07.2023	Bank of Russia liability in digital form. Enables on-chain monetary settlement without bank intermediaries.
Taxation	Tax Code: ch.25, art.214.1, ch.26.2	DFA income taxed under general rules. Loan interest reduces borrower's taxable base.

Summary

TrustLayer is an infrastructure protocol enabling businesses to raise direct financing from investors without a bank as mandatory intermediary. All terms are fixed in a smart contract, execution is automatic, and risks are structured through a tranche model.

For whom	Value	Key figures
Borrower	Access to financing at market rates without bank bureaucracy	Rate ~20% vs 27% at bank. Saving ~RUB 68,000 per million.
Tranche A investor	Protected income. Losses absorbed by lower tranches first.	12% net p.a. vs 8% bank deposit. Income +81%.
Tranche B investor	Enhanced yield with conscious second-level risk.	18% net p.a. Risk is transparent and numerically defined.
Platform	Fee and spread income without taking credit risk on balance sheet.	RUB 70,500 net per deal. At 50 deals/month - RUB 3.5M.
Government (tax)	Transparent system with full tax reporting.	RUB 227,350 taxes per deal. Fully declared.